

Paper A-01: Earth Unified Environmental Transport: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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May 2026

Abstract

This opening Earth-systems paper states the Part A modelling frame. It treats environmental transport as a CDFD/AFL hypothesis in which heat, water, nutrients, sediment, pollutants, and biological energy are represented as domain-specific fluxes Φ moving against constraints C through surfaces with responsiveness S and structural memory M_s . The release-local runtime outputs are internal diagnostics and tests that could break the mapping, not field evidence of climate, hydrological, ecological, or geological claims.

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV uses this Earth-systems opening paper to set the measurement boundary for the rest of Part A.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, and Λ denotes the Life Number where used [2]. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. Subscripts restrict coefficients to the named pathway, domain, or experiment.

2 Earth-System Transport Frame

Earth systems are coupled transport networks. Atmospheric circulation moves heat and moisture; oceans redistribute heat and salinity; rivers move water and sediment; soils regulate nutrients and organic matter; ecosystems route biomass and information through trophic structure. CDFD/AFL does not replace the standard equations in these fields. It

supplies a common audit layer for asking where flow rises, where capacity tightens, where responsiveness is lost, and where memory keeps a system from returning to its earlier state.

3 Candidate Variable Map

CDFD term	Earth-system reading	Example proxy
Φ	environmental drive or throughput	heat flux, runoff, nutrient load
C	active constraint or capacity burden	aridity, channel resistance, toxicity
S	responsiveness or buffering	biodiversity, soil structure, wetland storage
M_s	structural memory	degradation, hysteresis, legacy contamination
Ψ_s	operating ratio	balanced, constrained, or overloaded regime

This map is hypothesis-level until the proxies, scales, datasets, and failure conditions are specified for a given Earth-system question.

4 Part A Sequence

The Part A papers move from broad environmental transport into atmosphere, ocean, hydrology, soil, ecology, biodiversity, pollution, agriculture, sustainability, collapse, synthesis, plate tectonics, and volcanology. The sequence is ordered to start with transport structure, then add specific environmental constraints and failure modes.

5 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

6 Current Runtime Diagnostic

The diagnostic script `Part.E.Synthesis/supplementary/run_partiv_discovery.py` includes selected Earth-system adapter rows for climate and atmosphere, ocean transport, hydrological flow, biodiversity, and pollution. The accompanying network-cascade diagnostic checks finite-value behavior in an overload and memory-locking stress test. I use those rows as a runtime audit, not as a replacement for Earth-system data.

7 Falsification Targets

The Earth-systems framing weakens if a domain cannot define measurable proxies for Φ , C , S , and M_s , if the proposed operating ratio fails to predict overload or recovery behavior, or if a standard domain model explains the same data without any added value from the CDFD/AFL mapping.

8 Conclusion

Part A uses CDFD/AFL as a shared transport language for Earth systems. The framework is useful only when it sharpens measurable hypotheses and failure conditions. It remains separate from operational climate, hydrological, ecological, agricultural, or geological advice.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs` and `Part_E_Synthesis/figures/`.

9 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.